```python
import pandas as pd
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, MinMaxScaler,
LabelEncoder, OrdinalEncoder, OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.compose import ColumnTransformer
from sklearn.model_selection import train_test_split

df = pd.read_csv('HeartDiseaseUCI.csv', index_col=0)

num_variables = ["age", "trestbps", "chol", "thalach", "ca",
"oldpeak"]

def set_num(x):
    if x == 0:
        return 0
    else:
        return 1

df["target"] = df["num"].apply(set_num)

# 5. Séparer le dataset en un jeu d'entraînement/validation et un jeu
de test.
# On utilisera la fonction train_test_split.

X = df.drop(["target", "num"], axis=1)
y = df["target"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size =
0.2, stratify = df["target"])

num_pipeline = Pipeline(steps = [
    ("imputer", SimpleImputer()),
    ("normalization", MinMaxScaler())
])

preprocessor = ColumnTransformer(transformers=
     [
        ("numeric", num_pipeline, num_variables)
     ]
)

preprocessor
```

```
ColumnTransformer(transformers=[('numeric',
                                 Pipeline(steps=[('imputer',
SimpleImputer()),
                                                 ('normalization',
                                                  MinMaxScaler())]),
                                 ['age', 'trestbps', 'chol',
'thalach', 'ca',
                                  'oldpeak'])])
```

```
X_train_clean = preprocessor.fit_transform(X_train)

X_test_clean = preprocessor.transform(X_test)

pd.DataFrame(X_train_clean).to_csv("./X_train_clean.csv", index=False)
pd.DataFrame(X_test_clean).to_csv("./X_test_clean.csv", index=False)
pd.DataFrame(y_train).to_csv("./y_train.csv", index=False)
pd.DataFrame(y_test).to_csv("./y_test.csv", index=False)

X_train = pd.read_csv('X_train_clean.csv')
X_test = pd.read_csv('X_test_clean.csv')
y_train = pd.read_csv('y_train.csv')
y_test = pd.read_csv('y_test.csv')
```